AUGMENTED DECISION

Gaussian regression and variable selection

From sparse estimation to choosing useful variables

Lecture 3 / Three hours, including a 15-minute break

A theoretical foundation followed by a 60-minute Python application.

Regression under squared loss

Assume $\mathbb{E}Y^2 < \infty$. Define $m_P(x) = \mathbb{E}[Y \mid X = x]$ and

$$R_P(f) = \mathbb{E}[(Y - f(X))^2].$$

Conditioning on X gives the orthogonal decomposition

$$R_P(f) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathbb{E}[(f(X) - m_P(X))^2].$$

Thus m_P is a Bayes predictor and

$$R_P(f) - R_P(m_P) = \|f - m_P\|^2_{L_2(P_X)}.$$

P_X is the marginal law of X. The squared $L_2(P_X)$ norm means expectation over a fresh covariate.

The fixed-design Gaussian linear regression

$$\mathbf{Y} = \mathbf{X}\beta^* + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n), \quad \beta^* \in \mathbb{R}^p.$$

- $\mathbf{X} \in \mathbb{R}^{n \times p}$ is fixed and known, with rows $x_i^\top$.
- $\mathbf{Y} \in \mathbb{R}^n$ is observed. The unknown parameter is β^*.
- $\sigma > 0$ is known for the theoretical tuning formulas.
- $G = \mathbf{X}^\top \mathbf{X}/n$ and $r = \mathrm{rank}(\mathbf{X})$.

All expectations and probabilities in this experiment condition on $\mathbf{X}$. No unpenalized intercept is included; any such extension needs separate treatment.

Coefficient error and prediction error

For $\Delta = \widehat{\beta} - \beta^*$, define the realized estimation losses

$$\mathcal{L}_2 = \|\Delta\|_2^2, \qquad \mathcal{L}_{\mathbf{X}} = \frac{1}{n}\|\mathbf{X}\Delta\|_2^2 = \Delta^\top G \Delta.$$

Their expected risks are $\mathcal{R}_2 = \mathbb{E}_{\beta^*}\mathcal{L}_2$ and $\mathcal{R}_{\mathbf{X}} = \mathbb{E}_{\beta^*}\mathcal{L}_{\mathbf{X}}$.

- $\mathcal{L}_2$ measures recovery of coefficients.
- $\mathcal{L}_{\mathbf{X}}$ measures recovery of the mean at the observed design points.
- If $G = I_p$, the two losses coincide. In general, they do not.

Connection to future-response prediction

For independent new noise at the same design, $\mathbf{Y}^{\mathrm{new}} = \mathbf{X}\beta^* + \varepsilon^{\mathrm{new}}$,

$$\mathbb{E}\left[\frac{1}{n}\left\|\mathbf{Y}^{\mathrm{new}} - \mathbf{X}\widehat{\beta}\right\|_2^2 \,\middle|\, \mathbf{Y}\right] = \sigma^2 + \mathcal{L}_{\mathbf{X}}.$$

For a fresh covariate with second-moment matrix $\Sigma_{\mathrm{test}} = \mathbb{E}[XX^\top]$,

$$\text{excess prediction loss} = \Delta^\top \Sigma_{\mathrm{test}} \Delta.$$

A fixed-design bound controls future covariates only when their geometry is related to G.

Least squares as an orthogonal projection

$$\widehat{\beta}^{\mathrm{LS}} \in \operatorname*{arg\,min}_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

If $\mathrm{rank}(\mathbf{X}) = p$,

$$\widehat{\beta}^{\mathrm{LS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For any rank, the fitted mean is unique:

$$\widehat{\mu}^{\mathrm{LS}} = P_{\mathbf{X}} \mathbf{Y}, \qquad P_{\mathbf{X}} \text{ projects onto } \mathrm{col}(\mathbf{X}).$$

When rank is deficient, coefficient minimizers need not be unique.

Exact least-squares prediction risk

Let $\mu^* = \mathbf{X}\beta^*$. Since $P_{\mathbf{X}}\mu^* = \mu^*$,

$$\widehat{\mu}^{\mathrm{LS}} - \mu^* = P_{\mathbf{X}}\varepsilon.$$

Using $P_{\mathbf{X}}^2 = P_{\mathbf{X}}$ and $\mathrm{tr}(P_{\mathbf{X}}) = r$,

$$\mathcal{R}_{\mathbf{X}}(\widehat{\beta}^{\mathrm{LS}}, \beta^*) = \frac{\sigma^2}{n}\,\mathrm{tr}(P_{\mathbf{X}}) = \frac{\sigma^2 r}{n}.$$

In fact,

$$\frac{n\mathcal{L}_{\mathbf{X}}}{\sigma^2} \sim \chi^2_r.$$

Exact for every β^* in the correctly specified Gaussian model.

Exact coefficient risk and nonidentifiability

If $\mathrm{rank}(\mathbf{X}) = p$, then

$$\widehat{\beta}^{\mathrm{LS}} - \beta^* \sim \mathcal{N}\left(0, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}\right),$$

$$\mathcal{R}_2(\widehat{\beta}^{\mathrm{LS}}, \beta^*) = \sigma^2 \,\mathrm{tr}((\mathbf{X}^\top \mathbf{X})^{-1}).$$

If $u \neq 0$ lies in $\ker(\mathbf{X})$, β^* and $\beta^* + tu$ induce the same data distribution for every t.

Unrestricted coefficient recovery has infinite worst-case risk under rank deficiency.

A minimax risk must specify the loss

For a parameter set $\mathcal{B} \subseteq \mathbb{R}^p$ and loss $\mathcal{L}$, define

$$\mathfrak{M}(\mathcal{B}, \mathcal{L}; \mathbf{X}) = \inf_{\widehat{\beta}} \sup_{\beta^* \in \mathcal{B}} \mathbb{E}_{\beta^*} \mathcal{L}(\widehat{\beta}, \beta^*).$$

- The infimum covers all measurable estimators based on $(\mathbf{X}, \mathbf{Y})$.
- The design, noise level, parameter set, and loss are fixed in this definition.
- An upper bound comes from an estimator. A lower bound constrains every estimator.

For mean prediction, one may equivalently estimate $\mu^* \in \mathrm{col}(\mathbf{X})$. Projection onto that space cannot increase squared loss.

Exact Gaussian minimax benchmarks

For known $\sigma > 0$ and fixed design of rank r,

$$\inf_{\widehat{\mu}} \sup_{\mu \in \mathrm{col}(\mathbf{X})} \mathbb{E}_{\mu} \frac{1}{n} \|\widehat{\mu} - \mu\|_2^2 = \frac{\sigma^2 r}{n}.$$

If $\mathbf{X}$ has full column rank,

$$\inf_{\widehat{\beta}} \sup_{\beta \in \mathbb{R}^p} \mathbb{E}_{\beta} \left\|\widehat{\beta} - \beta\right\|_2^2 = \sigma^2 \operatorname{tr}((\mathbf{X}^\top \mathbf{X})^{-1}).$$

Least squares attains both values under the corresponding rank assumptions.

These are exact unbounded-parameter minimax risks, not sparse-model rates.

Proof: a diffuse Gaussian prior gives the lower bound

Choose an orthonormal basis $U \in \mathbb{R}^{n \times r}$ of the mean space. Write

$$\mu = U\theta, \qquad U^\top \mathbf{Y} = \theta + \sigma\xi, \quad \xi \sim \mathcal{N}(0, I_r).$$

Under the proof prior $\theta \sim \mathcal{N}(0, a^2 I_r)$, posterior variance in each coordinate is

$$v_a = \left(a^{-2} + \sigma^{-2}\right)^{-1} = \frac{a^2\sigma^2}{a^2 + \sigma^2}.$$

Worst-case risk is at least optimal prior-average risk, hence

$$\mathfrak{M} \geq \frac{r v_a}{n} \longrightarrow \frac{r\sigma^2}{n} \quad \text{as } a \to \infty.$$

The Gaussian sequence model

If $G = I_p$, then necessarily $p \leq n$ and

$$W = \frac{1}{n}\mathbf{X}^\top \mathbf{Y} = \beta^* + \tau\xi, \qquad \tau^2 = \frac{\sigma^2}{n}, \quad \xi \sim \mathcal{N}(0, I_p).$$

Under this orthogonal design,

$$\mathcal{L}_{\mathbf{X}} = \mathcal{L}_2, \qquad \inf_{\widehat{\beta}} \sup_{\beta \in \mathbb{R}^p} \mathbb{E}_\beta \mathcal{L}_2 = p\tau^2.$$

The sequence model is also a standalone experiment. It does not assert that a design with $p > n$ can have orthogonal columns. [RH, Section 9]

Sparsity changes the parameter set

Define $\|\beta\|_0 = |\operatorname{supp}(\beta)|$ and

$$\mathcal{B}_s = \{\beta \in \mathbb{R}^p : \|\beta\|_0 \le s\}, \qquad 1 \le s \le p.$$

If a support J of size s were known in the sequence model,

$$\widehat{\beta}_j = W_j \ (j \in J), \qquad \widehat{\beta}_j = 0 \ (j \notin J)$$

has worst-case coefficient risk $s\tau^2$ on that subspace.

With unknown support, the estimator must also discover where the signal lies.

Sparse Gaussian minimax rate

In $W = \beta^* + \tau\xi$, let

$$\mathfrak{M}_s = \inf_{\widehat{\beta}} \sup_{\beta^* \in \mathcal{B}_s} \mathbb{E}_{\beta^*} \left\| \widehat{\beta} - \beta^* \right\|_2^2.$$

Universal $c, C > 0$ satisfy

$$c\tau^2 s \log(ep/s) \leq \mathfrak{M}_s \leq C\tau^2 s \log(ep/s).$$

- Known support costs $s\tau^2$.
- Unknown support adds a logarithmic search cost.
- At $s = p$, the scale reduces to $p\tau^2$.

Theorem for the Gaussian sequence experiment. [RH, Section 19.2]

An upper bound from best subset selection

In the sequence model, define the ideal sparse estimator

$$\widehat{\beta}^{(s)} \in \operatorname*{arg\,min}_{\|b\|_0 \le s} \|W - b\|_2^2 .$$

It retains the s largest absolute coordinates of W and sets the others to zero.

Optimality and Gaussian maximal inequalities give

$$\sup_{\beta^* \in \mathcal{B}_s} \mathbb{E} \left\| \widehat{\beta}^{(s)} - \beta^* \right\|_2^2 \le C\tau^2 s \log(ep/s).$$

The support size s is supplied here. General correlated-design subset selection is computationally harder.

Variable selection has its own target

Let $S = \mathrm{supp}(\beta^*) = \{j : \beta^*_j \neq 0\}$ and $s = |S|$. A selection procedure returns $\widehat{S} \subseteq \{1, \dots, p\}$.

$$\mathrm{FP} = |\widehat{S} \setminus S|, \qquad \mathrm{FN} = |S \setminus \widehat{S}|.$$

Exact support recovery means $\widehat{S} = S$. Its probability, $\mathbb{P}_{\beta^*}(\widehat{S} = S)$, is a different performance criterion from squared prediction loss.

Example: $S = \{1, 3, 7\}$ and $\widehat{S} = \{1, 3, 9\}$ give one false positive and one false negative.

A useful prediction can coexist with an incorrect explanation of which coefficients are nonzero.

Why applications need a variable shortlist

- A warehouse team may want fewer measurements for forecasting processing time.
- An engineering team may prioritize sensors for a follow-up investigation.
- A reporting system may need a compact model that users can inspect.

Three questions remain distinct:

Prediction	Which inputs give low error on new observations?
Support recovery	Which coefficients are nonzero in the stated model?
Causal intervention	Which changes would alter the outcome?

Regression and variable selection alone do not identify the effect of an intervention.

Correlated variables can be hard to distinguish

If two columns are identical, a signal on either column produces the same mean.

$$\mathbf{X}_1 = \mathbf{X}_2 \quad \Longrightarrow \quad \mathbf{X}(a, 0, \ldots)^\top = \mathbf{X}(0, a, \ldots)^\top.$$

- Prediction may be accurate while the selected variables are ambiguous.
- Exact support recovery needs further assumptions, including signal strength and suitable design structure.
- A nonzero fitted coefficient does not establish a causal effect.

A simple support-recovery guarantee

In the orthogonal sequence experiment $W = \beta^* + \tau\xi$, take

$$t = \tau\sqrt{2\log(2p/\alpha)}, \qquad \widehat{S} = \{j : |W_j| > t\}, \quad 0 < \alpha < 1.$$

The Gaussian union bound gives $\mathbb{P}(\max_j |W_j - \beta_j^*| \le t) \ge 1 - \alpha$. If $S \neq \varnothing$ and

$$\beta_{\min} := \min_{j \in S} |\beta_j^*| > 2t, \qquad \text{then} \qquad \mathbb{P}(\widehat{S} = S) \ge 1 - \alpha.$$

On that event, null coordinates are at most t and signal coordinates exceed t. This is a sufficient condition for this orthogonal model, not a claim about every correlated design.

Forward selection fits ordinary linear models

Start with the intercept-only model, $J_0 = \varnothing$. At step k, try each unused variable j:

$$j_k \in \arg\min_{j \notin J_{k-1}} \min_{b_0, b} \sum_{i \in \text{train}} \left(Y_i - b_0 - x_{i, J_{k-1} \cup \{j\}}^\top b \right)^2, \quad J_k = J_{k-1} \cup \{j_k\}.$$

- Build the path $J_0, \ldots, J_8$ using training data only.
- Choose k by the smallest validation MSE along that path.
- Save the selected variables and their fitted coefficients.

This is a greedy search. It is not exhaustive best-subset selection and has no automatic exact-support guarantee. Training error cannot choose when to stop.

An intercept and training-only scaling

The application includes an unpenalized intercept:

$$Y_i = b_0 + x_i^\top \beta + \varepsilon_i.$$

Compute feature means $\bar{x}_j$, scales s_j, and response mean $\bar{y}$ using training rows only. Put $\tilde{x}_{ij} = (x_{ij} - \bar{x}_j)/s_j$.

$$\hat{\theta} = \arg\min_{\theta} \| Y - \bar{y}\mathbf{1} - \widetilde{X}\theta \|_2^2, \qquad \hat{y}(x) = \bar{y} + \sum_j \frac{x_j - \bar{x}_j}{s_j} \hat{\theta}_j.$$

Use `numpy.linalg.lstsq`, not an explicit matrix inverse. Apply the same training transformation to validation and test rows.

Theoretical no-intercept formulas above require the stated adjustment; see the appendix.

Shared case: forecasting warehouse completion time

Each row is a synthetic warehouse batch. Predict `completion_minutes` using only measurements available before processing starts.

- 24 continuous candidate indices, with unequal scales and correlated pairs.
- 60 training rows, 60 validation rows, and 120 held-out test feature rows.
- The test outcomes and generating coefficients stay with the instructor until the end of Lecture 4.

Recommend a forecasting model and a variable shortlist, using prediction error and the stability of the selected variables.

This is a deliberately small, noisy Gaussian linear example. The target, feature set, split, and evaluation protocol stay fixed across both labs.

The lab files and the handoff to Lecture 4

Open `lecture_03_student.ipynb` in the supplied Python lab folder.

- Fit an intercept-only baseline and OLS using all 24 variables.
- Run forward selection and choose its path length with validation data.
- Record training/validation MSE and your selected variables.
- Save `lecture3_handoff.json` and a short recommendation.

Do not open the test labels or generating truth. Keep the Lecture 3 fits for the comparison in Lecture 4.

15-minute break

We resume at elapsed time 02:00.

Next: the 60-minute variable-selection lab.

Lab 1: OLS and a variable shortlist

0–10 min	Load the fixed split, inspect the data, and fit training-only scaling.
10–25 min	Implement OLS with an intercept and evaluate the baseline and full model.
25–45 min	Build the training-only forward path up to eight variables.
45–55 min	Select its length by validation MSE and interpret the shortlist.
55–60 min	Save the handoff and state one limitation of the selected variables.

Deliverables: executable OLS code, an error table, selected feature names, and a handoff file. The next lecture compares these fits with ridge and lasso on the same task.

Appendix: coefficient minimax lower bound

Suppose $\mathbf{X}^\top \mathbf{X}$ is invertible and use the proof prior $\beta \sim \mathcal{N}(0, a^2 I_p)$. The posterior covariance is

$$V_a = (a^{-2} I_p + \sigma^{-2} \mathbf{X}^\top \mathbf{X})^{-1}.$$

Squared-error Bayes risk equals $\operatorname{tr}(V_a)$, so

$$\inf_{\widehat{\beta}} \sup_{\beta \in \mathbb{R}^p} \mathbb{E}_\beta \left\| \widehat{\beta} - \beta \right\|_2^2 \geq \lim_{a \to \infty} \operatorname{tr}(V_a) = \sigma^2 \operatorname{tr}((\mathbf{X}^\top \mathbf{X})^{-1}).$$

Least squares attains the bound. If the design is singular, a null-space prior variance grows without bound as $a \to \infty$.

Appendix: a Gaussian prediction deviation bound

For least squares, $n\mathcal{L}_{\mathbf{X}}/\sigma^2 \sim \chi^2_r$. For $t > 0$, the Gaussian quadratic-form Chernoff bound gives

$$\mathbb{P}\{\chi^2_r > r + 2\sqrt{rt} + 2t\} \leq e^{-t}.$$

Consequently, with probability at least $1 - \alpha$,

$$\mathcal{L}_{\mathbf{X}} \leq \frac{\sigma^2}{n}\left[r + 2\sqrt{r\log(1/\alpha)} + 2\log(1/\alpha)\right].$$

This controls realized mean prediction loss. Its expected value is exactly $\sigma^2 r/n$.

Appendix: sparse Fano lower bound

For $s \le p/8$, choose binary vectors $u^1, \dots, u^M$, each with exactly s ones, with Hamming distances at least cs and $\log M \ge c's \log(ep/s)$. Take $\beta^j = au^j$ and reference $\beta^0 = 0$. Then

$$\|\beta^j - \beta^k\|_2^2 \ge csa^2, \qquad \mathrm{KL}(P_{\beta^j} \| P_0) = \frac{sa^2}{2\tau^2}.$$

If a^2 is a sufficiently small multiple of $\tau^2 \log(ep/s)$, Fano gives a testing error bounded below by a positive constant:

$$\mathbb{P}(\text{error}) \ge 1 - \frac{M^{-1} \sum_j \mathrm{KL}(P_{\beta^j} \| P_0) + \log 2}{\log M}.$$

Nearest-packing-point decoding transfers this to squared estimation loss.

Small packing sizes use a two-point bound. For $s > p/8$, a fixed-support Gaussian lower bound gives the same rate up to constants. [RH, Section 19.2]

Appendix: the best-subset upper bound

Let $\Delta = \widehat{\beta}^{(s)} - \beta^*$ and $k = \min(2s, p)$. Optimality gives

$$\|\Delta\|_2^2 \le 2\tau \langle \xi, \Delta \rangle \le 2\tau \max_{|J| \le k} \|\xi_J\|_2 \|\Delta\|_2 ,$$

so $\|\Delta\|_2^2 \le 4\tau^2 \max_{|J| \le k} \|\xi_J\|_2^2$. A union bound over supports and the chi-square tail yield

$$\mathbb{E} \max_{|J| \le k} \|\xi_J\|_2^2 \le Ck \log(ep/k) \le C' s \log(ep/s).$$

This bounds expected risk uniformly over every amplitude of the sparse signal.

Appendix: transferring sparse rates to a design

A useful sufficient design assumption is a $2s$-sparse isometry:

$$a\,\|u\|_2^2 \le \frac{1}{n}\,\|\mathbf{X}u\|_2^2 \le b\,\|u\|_2^2 \quad \text{whenever } \|u\|_0 \le 2s, \quad 0 < a \le b < \infty.$$

Under fixed a, b, Gaussian noise, and $1 \le s \le p/8$, sparse minimax coefficient and mean-prediction risks have scale

$$\frac{\sigma^2 s \log(ep/s)}{n}.$$

Lower bounds use sparse packing and $\mathrm{KL}(P_\beta \| P_{\beta'}) = \|\mathbf{X}(\beta - \beta')\|_2^2 / (2\sigma^2)$. Upper bounds use ideal constrained least squares and union bounds over sparse subspaces.

Constants depend on a, b. This is an additional design-specific benchmark, not a consequence of Gaussian noise alone. See [RWY].

Appendix: the intercept adjustment

Let $C = I_n - \mathbf{1}\mathbf{1}^\top/n$, $X_c = CX$, and $Y_c = CY$. For the raw model $Y = b_0^*\mathbf{1} + X\beta^* + \varepsilon$,

$$Y_c = X_c\beta^* + C\varepsilon, \qquad \mathrm{Cov}(C\varepsilon) = \sigma^2 C.$$

The centered noise is not independent across its n coordinates. Still, $X_c^\top C = X_c^\top$, so full-rank slope OLS has covariance $\sigma^2(X_c^\top X_c)^{-1}$.

$$\text{Full fitted-mean risk} = \frac{\sigma^2}{n}\,\mathrm{rank}[\mathbf{1}, X].$$

For the lab, the design with an intercept has rank 25. Forward selection fitted on chosen variables does not inherit an unconditional fixed-subspace risk formula after data-dependent selection.

References for regression and sparse estimation

- **[RH]** Rigollet and Hütter, *High-Dimensional Statistics*, Sections 8–10 and 15–19.
  Author notes, arXiv:2310.19244
- **[RT]** Ryan Tibshirani, *Minimax theory*, *Ridge*, and *Lasso*, Berkeley Statistical Learning notes (2023).
  Course notes and sources
- **[RWY]** Raskutti, Wainwright, and Yu (2011), "Minimax rates of estimation for high-dimensional linear regression over ℓ_q-balls."
  Author manuscript, arXiv:0910.2042

The exact Gaussian risks use the experiment and loss definitions stated here. Ridge and lasso follow in Lecture 4.